

# File permissions in Linux

## Project description

As a security professional at a large organization, I am examining file system permissions for the members of the research team. My task is to review current permissions, determine unauthorized access, and modify the permissions to ensure proper authorization.

## Check file and directory details

```
researcher2@df5614720b17:~$ pwd
/home/researcher2
researcher2@df5614720b17:~$ ls
projects
researcher2@df5614720b17:~$ cd projects
researcher2@df5614720b17:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  2 16:28 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun  2 16:58 ..
-rw--w---- 1 researcher2 research_team  46 Jun  2 16:28 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun  2 16:28 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun  2 16:28 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun  2 16:28 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 16:28 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 16:28 project_t.txt
researcher2@df5614720b17:~/projects$
```

To determine my current location in the file system, I used the `pwd` command to display the full directory path. After that, I used the `ls` command to view the contents of the current path, which showed the `projects` directory. After navigating to the `projects` directory with `cd` command, I used `ls -la` to display a detailed listing of all files and directories, including hidden files. This command showed that the `projects` directory contains a directory named `drafts`, a hidden file named `.project_x.txt`, and five other files. The permissions for each item are displayed as a 10-character string in the first column of the output.

## Describe the permissions string

The 10-character string represents the file type and the access permissions for a file or directory. It helps determine who can read, modify, or execute in the file system:

- The first character shows the file type: `d` represents a directory / `-` represents a regular file.
- The first three characters show the Owner permissions.
- The middle three characters show the Group permissions.
- The last three characters show Others permissions.
- For each group, `r` stands for read, `w` stands for write, and `x` stands for execute. A dash `(-)` means that the specific permission is not authorized.

## Change file permissions

```
researcher2@ff9ff079beb9:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun  2 17:18 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun  2 17:18 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun  2 17:18 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 17:18 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 17:18 project_t.txt
researcher2@ff9ff079beb9:~/projects$ chmod o-w project_k.txt
researcher2@ff9ff079beb9:~/projects$ chmod g-r project_m.txt
researcher2@ff9ff079beb9:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun  2 17:18 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 17:18 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun  2 17:18 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 17:18 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 17:18 project_t.txt
researcher2@ff9ff079beb9:~/projects$
```

While reviewing the file system permissions, I identified that Others had write access on the `project_k.txt` file, and Group had read access on the restricted file `project_m.txt`. This level of access was not appropriate based on the security requirements. I used the `chmod` command to remove write access for Others and read access for Group. After making the change, I used `ls -l` to verify the updated permissions. The output confirmed that write access for the Others group had been successfully removed from the `project_k.txt` file, while read access for the Group was removed from `project_m.txt`.

## Change file permissions on a hidden file

```
researcher2@ff9ff079beb9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  2 17:18 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun  2 18:09 ..
-rw--w---- 1 researcher2 research_team  46 Jun  2 17:18 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun  2 17:18 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 17:18 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun  2 17:18 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 17:18 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 17:18 project_t.txt
researcher2@ff9ff079beb9:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@ff9ff079beb9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  2 17:18 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun  2 18:09 ..
-r--r----- 1 researcher2 research_team  46 Jun  2 17:18 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun  2 17:18 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 17:18 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun  2 17:18 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 17:18 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 17:18 project_t.txt
researcher2@ff9ff079beb9:~/projects$
```

To view hidden files in the directory, I used the `ls -la` command. From the output, I identified the hidden file `.project_x.txt`. The requirements specified that no one should have write access to this file, but the Owner and Group should have read access. To apply these changes, I used the `chmod` command to remove write permissions from Owner and Group, while ensuring read permissions were added to the Group. After making the changes, I used `ls -la` again to verify that the updated permissions were applied successfully.

## Change directory permissions

```
researcher2@f8d7d8d58df8:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun  2 18:56 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun  2 18:56 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun  2 18:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 18:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 18:56 project_t.txt
researcher2@f8d7d8d58df8:~/projects$ chmod g-x drafts
researcher2@f8d7d8d58df8:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Jun  2 18:56 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun  2 18:56 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun  2 18:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 18:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun  2 18:56 project_t.txt
researcher2@f8d7d8d58df8:~/projects$ █
```

To review directory permissions, I used the `ls -l` command. Based on the requirements, only the Owner should have full permissions (read, write, and execute) for the `drafts` directory. To authorize proper permissions, I used the `chmod` command to remove execute permissions from the Group. After applying the change, I used `ls -l` again to verify that the updated permissions were correctly applied.

## Summary

In this activity, I used Linux commands to manage file and directory permissions of the `projects` directory. I began by using `ls -la` to review existing permissions and identify hidden files. After that, I used `chmod` to remove unauthorized access and adjust permissions on hidden files, regular files, and directories. These changes ensured that all users had appropriate access, improving the security of the file system.